CHAPTER 12

The Origin of the Universe and Nucleosynthesis

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Today it is assumed that our universe consists of ~70% dark energy, ~25% dark matter, and ~5% of radiation and visible matter, see Table 12.1. Hydrogen, deuterium and most of the helium atoms in the normal matter part of the universe are believed to have been created 13.75 ± 0.11 billion years ago in a primary formation process referred to as the *Big Bang*, while all other elements have been formed — and are still being formed — in nuclear reactions in the stars and today also in our nuclear reactors. Most of these reaction processes can only be understood in an astrophysical context, as briefly outlined in this chapter, which also describes how nuclear science has provided much understanding about the universe, our solar system and our planets. The fusion reactions, which created the lightest elements, may in the future also provide us with a large source of energy in controlled thermonuclear reactors, see §19.17. It follows from the understanding of the processes involved that both fission and fusion used for power producing purposes should be named *nuclear power.*

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Table 12.1 Composition of the universe

Constituent	% of Universe mass
Dark energy	70
Dark matter	25
Free H and He	4
Stars	0.5
Neutrinos	0.3
Elements heavier than He	0.03

12.1. OBSERVATIONS FROM SPACE PROBE EARTH

12.1.1. Our place in the universe

The stars we directly can see with our eyes all belong to our Galaxy, the *Milky Way*, which is a barred spiral galaxy, about 30 kpc (kpc = kiloparsec) across and about 1 kpc thick, See Fig.12.1. The parsec (or parallax second) is the common astronomical unit of distance; 1 parsec is 3.1×10^{16} m, or 3.26 light years, ly, which corresponds to a one second change in direction angle when the earth moves between two opposite positions around the sun. Thus light travels across our galaxy in about 100 000 y. The Milky Way contains some 200×10^9 stars, interstellar dust and gas ($\sim$200 pc thick) which spreads out to a diameter of about 50 kpc (hot gas atoms and single stars, the *halo*). It has two long spiral arms and a short one beginning from the ends of a central bar. Our sun is located in a spur from the outer edge of the short spiral arm, about 8.5 kpc from the galactic center as shown in Figures 12.1 and 12.2. The dust limits the sight at visual wave-lengths towards the center to only a few kpc; without this dust the Galaxy center would shine equally bright as our sun. Infrared- and radio-telescopes have made it possible to penetrate much of the dust and show stars circling very rapidly around the center. The stars in the bar seems at the same time to rotate like a cylinder. The other stars in our Galaxy move tangentially around its center with angular velocities also increasing closer to the center. Both observations indicate the existence of an extremely heavy central object, called Sagittarius (Sgr) A^*.

The Milky Way belongs to the *Local Group*, a cluster of some 20 galaxies which include, i.a. the Large Magellanic Cloud, our nearest galaxy, 50 kpc away and the Andromeda galaxy, 650 kpc away. The Local Group is part of the larger *Virgo super-cluster.* The universe contains some 10^{10} galaxies, but galaxies, stars and gases only fill a fraction of space, <5%, the rest appears void of matter. On a grand scale it looks like the galaxies and empty space form a bubbly structure, see Figure 12.3.

In the 1930s Hubble discovered that galaxies on the whole are equally distributed in all directions of space as observed from the earth. Thus space – on a large scale – seem to be isotropic. This idea of uniformity of the universe is called *the cosmological principle.*